HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Designed and implemented **Python** feature pipelines on **AWS Lambda** and **S3** to power downstream ML scoring, handling approximately **3,500 events/sec** during peak hours and replacing a brittle scheduled-batch job.
- Engineered a **Python** data extraction and transformation process for a large-scale financial transactions dataset, generating ~**1.1M records** for downstream analytics and model training.
- Developed and maintained **Java Spring Boot** REST APIs for cross-service data exchange, incorporating request validation and circuit-breaker patterns to enhance system resilience.
- Collaborated with quantitative researchers and platform engineers to align compute and storage capabilities, contributing to the architecture of the core ML platform infrastructure.
- Wrote and maintained **Python** scripts for data ingestion and ETL processes, automating report generation and reducing manual processing time by an estimated **40%**.
- Containerized backend services using **Docker** and deployed them to a **Kubernetes** cluster, managing Helm chart configuration and basic resource allocation for predictable rollouts.
- Supported production enterprise applications, responding to P1/P2 incidents and managing SLAs using **Splunk** and **Dynatrace** for monitoring and log analysis.

SKILLS

Languages – Python, Java, SQL, Bash, R

Frameworks & Libraries – Spring Boot, scikit-learn, pandas, NumPy, Polars, PyTorch, XGBoost, FastAPI

Distributed Systems & Compute – Ray, Dask, Spark, PySpark, Kubernetes, Docker, Arrow, Slurm

Databases & Data Storage – PostgreSQL, MySQL, Oracle, S3

Cloud & CI/CD – AWS (EC2, Lambda, CloudWatch), Jenkins, Git

Observability & Tools – Splunk, Dynatrace, Jira, Confluence, Unix/Linux

PROJECTS

ML Feature Pipeline for Transaction Analysis

- Engineered a **Python** feature pipeline to extract behavioural signals (velocity, geo-distance) from a transaction log of approximately **1.1M records**.
- Trained a **XGBoost** classifier on the extracted features, tuning class weights to handle imbalanced fraud data and achieving **~0.91 AUC** on a held-out month.
- Deployed the model behind a **FastAPI** inference endpoint, implementing input validation and monitoring p95 latency, which averaged **180ms**.

Distributed Dataset API Enhancements

- Extended a **Polars**-style distributed dataset API by implementing new operators and optimizations for lazy evaluation and DAG planning.
- Designed durable abstractions to ensure the platform's flexibility as distributed compute backends evolve, supporting multiple **Python**-based data processing workflows.
- Developed unit tests and documentation for new API components, ensuring adherence to strong engineering practices and facilitating easier adoption for internal researchers.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)